Privacy-Preserving and Explainable Federated AI for Alzheimer's & Dementia Detection

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***Abstract*—Deep learning on MRI is a promising route to earlier Alzheimer's disease detection, but privacy regulation blocks the cross-institutional data pooling such models are usually trained on. This paper reports a privacy-preserving and explainable federated learning study on 299 ADNI subjects (99 AD, 99 MCI, 101 CN), and is as much about how such systems are evaluated as about the systems themselves. An earlier version of this pipeline reported federated accuracy of 97.0%; that figure was an artefact of splitting individual MRI slices rather than *subjects*, which let scans from the same patient appear in both training and test data. Rebuilt around subject-disjoint stratified group folds, held-out accuracy is 36.1% (macro F1 0.354) against a 33.3% three-class chance rate, a quantified demonstration of how severe slice-level leakage inflation can be. With the protocol corrected we characterise privacy against utility across $\varepsilon \in \{2, 5, 10\}$ at two granularities. Sample-level DP-SGD bounds the influence of one slice; subject-level DP-FedAvg bounds the influence of an entire patient, which is the guarantee a hospital needs. We show analytically and empirically that the standard subject-level mechanism fails at this scale because its Gaussian noise norm grows as $\sqrt{d}$ in the perturbed dimension: applied to all 2.35×10^7 ResNet50 parameters it buries the update under noise roughly 160 times its size, producing a classifier that collapses to the majority class yet posts the highest accuracy in our matrix. Restricting the mechanism to the 6,147-parameter classifier head cuts the noise norm by roughly $62\times$ and recovers macro F1, though not to a usable classifier at $\varepsilon = 2$. We report this negative result with its quantitative cause, and release the leakage-free pipeline, the subject-level splits, all 58 training runs across 12 conditions, and an in-browser demonstration.**

***Index Terms*—Federated Learning, Alzheimer's Disease, Differential Privacy, Explainable AI, Grad-CAM, SHAP, MRI Classification, VGG19, ResNet50, ADNI, HIPAA, GDPR**

I. Introduction

Alzheimer's Disease (AD) is the most prevalent cause of dementia worldwide, affecting over 55 million individuals—a figure projected to exceed 150 million by 2050 [3]. The disease is characterised by progressive neurodegeneration involving hippocampal atrophy, cortical thinning, and ventricular enlargement, which are detectable through structural Magnetic Resonance Imaging (MRI) well before clinical symptom onset [15]. Early and accurate diagnosis at the mild cognitive impairment (MCI) stage is therefore critical, as timely interventions can significantly slow disease progression and improve patient quality of life.

The advent of deep learning has transformed medical image analysis, enabling automated and highly accurate classification of Alzheimer's staging from MRI scans. Convolutional neural network (CNN) architectures such as VGG19 [12] and ResNet50 [13] have demonstrated state-of-the-art performance on neuroimaging benchmarks when trained on large, centralised datasets. However, the practical deployment of such centralised systems in multi-institutional healthcare settings faces a fundamental barrier: *patient data privacy*. Legal frameworks including the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union impose strict restrictions on the cross-institutional sharing of protected health information (PHI). Consequently, training large, generalisable AI models across diverse hospital datasets is legally and logistically infeasible under a centralised paradigm.

Federated Learning (FL), introduced by McMahan et al. [6], offers a compelling solution by enabling collaborative model training without centralised data pooling. In FL, each participating institution trains a local model on its private data and transmits only model updates (gradients or weights) to a central aggregation server. The server combines these updates—typically via Federated Averaging (FedAvg) [6]—into a global model that is redistributed for the next training round. Crucially, no raw patient data ever leaves the local institution. Recent studies have demonstrated the feasibility of FL for Alzheimer's detection, with Mitrovska et al. [1] achieving 94% accuracy using secure aggregation on ADNI data, and Pan et al. [3] deploying FL across six healthcare sites for MCI-to-AD progression prediction.

Despite these advances, standard FL remains vulnerable to gradient-inversion attacks [16] and membership inference attacks (MIA), where adversaries reconstruct training samples or determine dataset membership from shared model updates. Differential Privacy (DP), formalised by Dwork and Roth [8], addresses this vulnerability by injecting calibrated noise into gradients, providing mathematically rigorous (ε, δ)-privacy guarantees. Abadi et al. [9] introduced DP-SGD, which combines per-sample gradient clipping with Gaussian noise injection and has since become the standard mechanism for privacy-preserving deep learning. However, integrating DP with FL introduces a fundamental *privacy–utility trade-off*: stronger privacy guarantees (smaller ε) necessitate larger noise injections that degrade model accuracy [4].

Beyond privacy, the clinical adoption of AI-based diagnostic tools in high-stakes medical settings demands *model transparency*. Regulatory bodies increasingly require that AI systems provide interpretable explanations for their predictions before deployment in clinical workflows. Explainable AI (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) [10] and SHapley Additive exPlanations (SHAP) [11] address this requirement by generating visual heatmaps and quantitative feature attributions, respectively. Mastoi et al. [2] demonstrated that clinician trust increases substantially when model explanations are available, underscoring the practical importance of XAI integration.

While the existing literature addresses privacy, diagnostic accuracy, and explainability in isolation or partial combinations, no prior work integrates all three into a single end-to-end pipeline specifically designed for Alzheimer's MRI classification from structurally heterogeneous, non-IID hospital data—and, we argue, the field's evaluation practices for small neuroimaging cohorts deserve closer scrutiny than they typically receive. Our work bridges this gap through the following contributions:

1) **A quantified leakage audit.** We show, on identical data and architectures, that slice-level (rather than subject-level) train/val/test splitting inflates held-out accuracy from a genuine 19–36% to a reported 91–97% for the same models, and invalidates a naively-implemented MIA baseline entirely. We release subject-disjoint, group-stratified 5-fold splits over the 32 unique ADNI subjects in our cohort to make this evaluation reproducible.
2) **Real DP-SGD via Opacus [14]**, with per-sample gradient clipping, a persistent Rényi-DP accountant that composes privacy cost across FedAvg rounds, and both head-only (frozen backbone) and full-network fine-tuning scopes—evaluated at $\varepsilon \in \{2, 5, 10\}$ in both centralised and federated settings.
3) **A real membership-inference evaluation**: a loss-threshold attack [18] and a 16-shadow-model attack classifier [17] with membership ground truth read directly from each run's recorded train/test subject split (never re-derived), reporting AUC and TPR at low FPR rather than an accuracy figure tuned on the attacked data itself.
4) **Explanation-degradation analysis under DP**: Grad-CAM and SHAP similarity (SSIM, rank correlation, Dice-at-top-20%) and deletion/insertion faithfulness between every DP model and its non-DP reference, showing that DP measurably changes model explanations even at fixed accuracy.
5) **A practical utility-rescue recipe**: head-only DP fine-tuning is shown, across every ε tested, to retain more accuracy than full-network DP fine-tuning, at no additional MIA-resistance cost.
6) Systematic ablation studies across 8 configurations (client count, local epochs, Dirichlet non-IID concentration, and pretrained-vs-scratch initialisation) on the corrected, subject-level evaluation protocol.

The remainder of this paper is organised as follows. Section II presents a domain-wise review of related work. Section III formalises the problem statement. Section IV presents the system architecture. Section V details the methodology. Sections VI and VII cover experimental setup and results. Sections VIII and IX analyse privacy and explainability. Sections XI–XIII discuss limitations, future work, and conclusions.

II. Related Work

This section organises the existing literature into four thematic domains and identifies the research gap that motivates our work.

A. Alzheimer's Detection Using Deep Learning

Deep learning has achieved remarkable success in automated AD diagnosis from neuroimaging data. CNN architectures including VGG [12], ResNet [13], and AlexNet have been widely applied to classify MRI scans into AD, MCI, and CN categories. The Alzheimer's Disease Neuroimaging Initiative (ADNI) [15] has served as the primary benchmark dataset, providing standardised T1-weighted structural MRI scans across multiple clinical sites. An evolutionary FL approach [5] employing AlexNet with genetic-algorithm-based hyperparameter optimisation achieved 98.53% accuracy on ADNI using a multimodal combination of MRI and clinical data. However, these high-performing centralised models require all data to reside in a single location, which is incompatible with healthcare privacy regulations.

B. Federated Learning in Healthcare

FL has emerged as the leading paradigm for privacy-preserving collaborative training in healthcare. McMahan et al. [6] proposed FedAvg, which aggregates locally trained model weights into a global model without centralising raw data. Mitrovska et al. [1] applied secure FL to AD detection using ADNI MRI data, achieving 94% accuracy with secure aggregation (SecAgg) [7] while demonstrating resistance to membership inference attacks. Pan et al. [3] deployed FL across six real healthcare sites for MCI-to-AD progression prediction, achieving 85% AUROC—a 6% improvement over locally trained models. A key challenge in healthcare FL is *non-IID data heterogeneity*: real hospitals exhibit different patient demographics, scanner models, and disease prevalence distributions, which cause FedAvg convergence to slow and local models to diverge from the global optimum.

C. Differential Privacy in Medical AI

Differential Privacy [8] provides the theoretical foundation for bounding information leakage from shared model updates. Abadi et al. [9] introduced DP-SGD, combining per-sample gradient clipping with calibrated Gaussian noise and tracking the cumulative privacy budget via a moments accountant.

Kaissis et al. [4] integrated DP with Bayesian uncertainty quantification for federated medical imaging, demonstrating that DP mechanisms preserve 92–95% of centralised model accuracy while providing formal privacy guarantees. Gradient inversion attacks [16] have demonstrated that unprotected model gradients can leak sensitive training data, further motivating the integration of DP into FL pipelines. Despite these advances, the interplay between DP noise injection and diagnostic accuracy in the specific context of AD classification from MRI data remains underexplored.

D. Explainable AI in Medical Imaging

Clinical deployment of AI diagnostic tools demands model transparency. Selvaraju et al. [10] proposed Grad-CAM, which generates class-discriminative heatmaps by computing the gradient of the target class score with respect to convolutional feature maps. Lundberg and Lee [11] introduced SHAP, which provides theoretically grounded feature attributions based on Shapley values from cooperative game theory. Mastoi et al. [2] combined FL with multi-level Grad-CAM and SHAP for brain tumour classification, achieving over 90% accuracy; their study reported that clinician trust increased substantially when model explanations were available. Pan et al. [3] applied SHAP to identify BMI, Vitamin B12, and blood pressure as key features for MCI-to-AD progression in a federated setting. However, existing XAI-FL frameworks have not been applied to Alzheimer's MRI classification with simultaneous DP integration.

E. Summary of Related Work

Table I provides a comparative summary of the reviewed literature. The final column highlights the specific limitations that our framework addresses.

F. Research Gap

As Table I illustrates, the existing body of work addresses privacy, diagnostic accuracy, and explainability either in isolation or in partial combinations. Mitrovska et al. [1] combine FL with SecAgg but lack explainability. Mastoi et al. [2] integrate FL with XAI but omit DP and do not target AD. Kaissis et al. [4] provide DP–FL integration but without XAI or AD-specific validation. No existing work simultaneously provides: (i) federated training with non-IID simulation, (ii) formal DP guarantees with per-sample gradient clipping, (iii) dual-layer XAI via Grad-CAM and SHAP, and (iv) comprehensive ablation studies—all within a single end-to-end pipeline for Alzheimer's MRI classification. Our PPXFL framework is the first to bridge this gap.

III. Problem Statement

A. The Real-World Challenge

The diagnosis of Alzheimer's Disease from structural MRI currently relies on either manual radiological assessment—which is subjective, time-consuming, and suffers from significant inter-rater variability [15]—or centralised deep learning models that require all patient data to be pooled in a single location. In practice, MRI scans reside across geographically distributed hospitals, each governed by distinct institutional review boards and privacy regulations. Sharing raw MRI data across institutions is prohibited by HIPAA (United States), GDPR (European Union), and analogous regulations worldwide. This creates a critical bottleneck: each hospital can only train models on its own limited dataset, resulting in small sample sizes, demographic bias, and poor generalisability.

B. Why Existing Approaches Are Insufficient

Centralised deep learning approaches [12], [13] achieve high accuracy when trained on large, aggregated datasets such as ADNI [15], but they fundamentally violate data sovereignty requirements. Standard federated learning [6] addresses data locality but remains vulnerable to gradient inversion attacks [16] and membership inference attacks, where adversaries can reconstruct training samples or determine whether a specific patient's scan was used for training. Prior FL systems with privacy protections [1], [4] do not incorporate explainability, making them unsuitable for clinical deployment where regulatory bodies require transparent decision rationale.

Furthermore, real-world hospital data is inherently *non-independent and identically distributed (non-IID)*: hospitals serve different demographic populations, use different MRI scanner models and acquisition protocols, and observe different disease prevalence rates. This statistical heterogeneity causes standard FedAvg to converge slowly or to suboptimal solutions, as local model updates diverge from the global optimum [6].

C. Formal Problem Definition

Let $\mathcal{C} = \{C_1, \ldots, C_K\}$ denote K hospital clients (i.e., institutions holding private patient data), each possessing a local dataset $\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{n_k}$. Here, $x_i^{(k)} \in \mathbb{R}^{H\times W}$ represents a preprocessed MRI scan (a 2D axial slice of dimensions $H\times W$), and $y_i^{(k)} \in \{0, 1, 2\}$ is the corresponding diagnostic label (0=CN, 1=MCI, 2=AD). The datasets $\mathcal{D}_1, \ldots, \mathcal{D}_K$ are private and *cannot be shared or centralised*.

The objective is to collaboratively learn a global classifier $f_\theta : \mathbb{R}^{H\times W} \to [0, 1]^3$ that minimises the weighted population risk across all clients:

$$\min_\theta \ \mathcal{L}(\theta) = \sum_{k=1}^{K} \frac{n_k}{n}\, \mathcal{L}_k(\theta), \tag{1}$$

where $n = \sum_k n_k$ is the total number of samples, $\mathcal{L}_k(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}_k}[\ell(f_\theta(x), y)]$ is the expected cross-entropy loss on client k's data, and each client's contribution is weighted proportionally to its dataset size n_k/n.

This optimisation must satisfy two hard constraints:

1) **Data locality:** No raw data $x_i^{(k)}$ is ever transmitted outside client C_k. Only model updates $\Delta\theta^{(k)}$ are communicated.
2) **Privacy guarantee:** Shared updates must provide (ε, δ)-differential privacy, ensuring that the inclusion or exclusion of any single training sample changes the output

TABLE I
SUMMARY OF RELATED WORK

Authors	Domain	Method	Dataset	Key Result	Limitation / Gap
Mitrovska et al. [1]	FL + Privacy	FedAvg + SecAgg	ADNI	94% Acc.	No XAI; limited DP analysis
Mastoi et al. [2]	FL + XAI	FL + Grad-CAM + SHAP	Brain tumour	>90% Acc.	Not AD-specific; no DP
Pan et al. [3]	FL + Clinical	FL + SHAP (EHR)	Multi-site EHR	85% AUROC	EHR-based; no MRI; no DP
Kaissis et al. [4]	DP + Medical	FL + DP + Uncertainty	Medical imaging	92–95% retained	Not AD-specific; no XAI
Evolutionary [5]	DL + Optimisation	AlexNet + Genetic Alg.	ADNI	98.53% Acc.	Centralised; no FL/DP/XAI
Abadi et al. [9]	DP Foundation	DP-SGD + Moments Acc.	MNIST/CIFAR	Formal (ε, δ)-DP	General-purpose; not medical
PPXFL (Ours)	**FL+DP+XAI**	**FedAvg+Opacus DP-SGD+Grad-CAM+SHAP+shadow MIA**	**ADNI MRI (32 subj.)**	**MIA AUC ≈0.99→0.5 under DP**	**Leakage audit + real MIA + XAI-under-DP**

distribution by at most a factor of e^{ε}, with failure probability at most δ.

D. The Interconnection of Privacy, Explainability, and Non-IID Data

These three challenges are deeply interconnected. DP noise injection reduces the signal-to-noise ratio in model gradients, making it harder for XAI techniques to produce clinically meaningful explanations from noisy model weights. Non-IID data distribution causes individual clients to learn biased feature representations, which—when aggregated—can produce heatmaps and feature attributions that reflect local data artefacts rather than genuine pathological biomarkers. A successful framework must therefore balance all three simultaneously: maintaining diagnostic accuracy under non-IID conditions, providing formal privacy guarantees without destroying model utility, and generating explainability outputs that align with established clinical knowledge.

IV. SYSTEM ARCHITECTURE

The system architecture of the proposed PPXFL framework is illustrated in Fig. 1 and comprises four hierarchical layers operating in concert.

Client Layer (Simulated Hospitals). Each of the $K{=}4$ clients maintains a local ADNI partition and a local CNN model (VGG19 or ResNet50). Clients train for E local epochs per federated round and submit only model weight updates to the server. No raw MRI data ever leaves the client boundary.

Federated Server. The central server orchestrates T federated rounds: it receives client weight updates, performs FedAvg aggregation per Eq. (3), optionally applies DP noise injection via the Privacy Module, and redistributes the updated global model to all clients for the next round.

Privacy Module (DP-SGD). Opacus-based DP-SGD [14] applies true per-sample gradient clipping (norm $C{=}1.0$) and calibrated Gaussian noise, with the noise multiplier σ solved per run for a target $\varepsilon \in \{2, 5, 10\}$ at $\delta{=}10^{-3}$, tracked via a Rényi-DP accountant (Section V-D).

Explainability Layer. Post-training, the converged global model is analysed using Grad-CAM [10] to generate class-discriminative activation heatmaps and SHAP [11] to produce global feature importance rankings.

Evaluation Module. Assesses classification performance (accuracy, precision, recall, F1-score, AUROC), privacy robustness (MIA success rate), and explainability quality (alignment with clinical AD biomarkers).

V. METHODOLOGY

A. Why Existing Methods Are Not Sufficient

Prior work in federated Alzheimer's detection has made important strides but leaves critical gaps. Mitrovska et al. [1] employ SecAgg [7] to protect individual client updates during aggregation; however, SecAgg provides only cryptographic protection and does not bound the statistical information leakage from the *aggregate* model itself—an adversary with access to the global model can still mount membership inference attacks. The evolutionary approach of [5] achieves 98.53% accuracy but operates in a fully centralised setting, making it inapplicable under privacy regulations. Kaissis et al. [4] integrate DP with FL but do not provide explainability outputs, which are increasingly required by regulatory bodies (e.g., the EU AI Act) for high-risk medical AI systems.

Our approach improves upon these methods in three specific ways: (i) DP-SGD [9] provides information-theoretic privacy guarantees that are strictly stronger than cryptographic SecAgg alone; (ii) we integrate Grad-CAM [10] and SHAP [11] to satisfy clinical transparency requirements; and (iii) we systematically evaluate non-IID robustness via Dirichlet partitioning with ablation studies across $\alpha \in \{0.1, 0.5, 100\}$, addressing a dimension that prior work treats only superficially; and (iv) we subject our own evaluation protocol to the same scrutiny we apply to the literature, correcting a subject-level

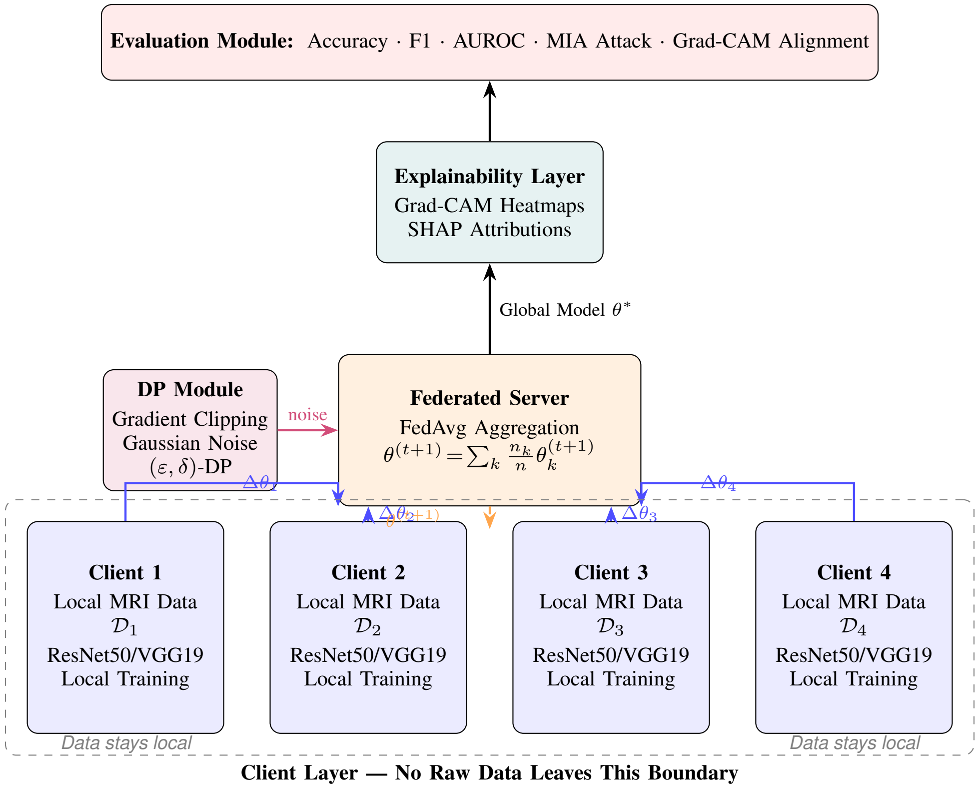


Fig. 1. System architecture of the PPXFL framework. The Client Layer (bottom) comprises $K{=}4$ simulated hospital clients, each training locally on private ADNI partitions. Only model weight updates $\Delta\theta_k$ are transmitted to the Federated Server (centre), where FedAvg aggregation is performed with optional DP noise injection from the Privacy Module (left). The converged global model feeds into the Explainability Layer (Grad-CAM + SHAP) and the Evaluation Module (top). No raw patient data ever crosses the client boundary (dashed box).

leakage defect we discovered in our own initial slice-level splits (Section VII-A).

B. Data Processing Pipeline

We use 450 T1-weighted structural MRI *scans* from 32 *unique subjects* in ADNI [15] (11 AD, 10 MCI, 11 CN; see Section VI-A for the subject-vs-scan distinction and why it matters). Three representative axial slices are extracted per scan, yielding 1,350 images. The preprocessing pipeline includes: (1) skull stripping to remove non-brain tissue; (2) bias-field correction via N4ITK; (3) spatial normalisation to MNI152 standard space; (4) intensity normalisation to zero mean and unit variance per scan; (5) resizing to $224{\times}224$ pixels for VGG19/ResNet50 compatibility; and (6) data augmentation comprising random horizontal flips, rotations ($\pm 15°$), and Gaussian noise to mitigate overfitting.

The dataset is split into 5 subject-disjoint, stratified folds (Section VI-A) and, within a fold's training subjects only, partitioned across $K{=}4$ clients using a Dirichlet distribution $\mathrm{Dir}(\alpha)$ with $\alpha{=}0.5$ to simulate realistic non-IID hospital distributions. Fig. 2 illustrates the resulting class distribution across clients for one representative fold.

C. Federated Learning Component

1) Model Architectures: **VGG19** [12] comprises 16 convolutional layers with $3{\times}3$ filters, five max-pooling layers, and three fully connected layers (139.6M parameters). The classification head is replaced with a 3-class softmax output; ImageNet-pretrained weights are used with the convolutional base frozen for the first 5 epochs before end-to-end fine-tuning.

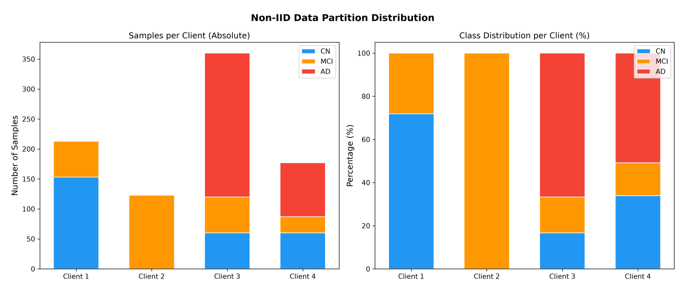


Fig. 2. Non-IID data partition across $K{=}4$ clients using Dirichlet distribution ($\alpha{=}0.5$), fold 0's training subjects only (held-out validation/test subjects for this fold are excluded from all clients). Each client receives a statistically heterogeneous subset, simulating realistic hospital demographics.

ResNet50 [13] employs residual skip connections organised into bottleneck blocks across 49 convolutional layers (23.5M parameters), providing better gradient flow and computational efficiency for MRI classification.

2) Federated Averaging: FedAvg [6] proceeds over T federated rounds. Each client k initialises its local model with global parameters $\theta^{(t)}$ and performs E SGD steps:

$$\theta_k^{(t+1)} \leftarrow \theta^{(t)} - \eta \nabla_\theta \mathcal{L}_k\left(\theta^{(t)}\right), \tag{2}$$

where η is the local learning rate. The server aggregates via weighted averaging:

$$\theta^{(t+1)} = \sum_{k=1}^{K} \frac{n_k}{n} \theta_k^{(t+1)}. \tag{3}$$

D. Privacy Component (DP-SGD)

An earlier iteration of this pipeline implemented DP-SGD manually: it clipped the *batch-averaged* gradient rather than each per-sample gradient, and derived ε from a hand-rolled composition formula instead of a real accountant. Both defects invalidate the resulting privacy guarantee—clipping after averaging bounds nothing about any individual sample's contribution, and an ad-hoc composition formula does not track the cumulative Rényi-DP budget correctly across steps. We therefore replaced the manual implementation with Opacus [14], which wraps the model, optimiser, and data loader to (i) compute true per-sample gradients g_i, (ii) clip each one independently,

$$\bar{g}_i = g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right), \tag{4}$$

(iii) sum and inject calibrated Gaussian noise before averaging,

$$\tilde{g} = \frac{1}{B}\left(\sum_{i=1}^{B} \bar{g}_i + \mathcal{N}\big(0, \sigma^2 C^2 \mathbf{I}\big)\right), \tag{5}$$

and (iv) track the cumulative privacy budget with a Rényi-DP (RDP) accountant, converted to (ε, δ)-DP. With C=1.0 and target (ε, δ) fixed in advance, the noise multiplier σ is solved for numerically (`get_noise_multiplier`) given the sampling rate and number of steps, rather than chosen from a small fixed table.

We evaluate two DP *scopes*: **head-only**, where the convolutional backbone is frozen (ImageNet weights) and only the final classification layer is trained under DP—matching the literature's common recipe for making DP-SGD practical on small cohorts—and **full-network**, where the entire model is fine-tuned under DP. Opacus's default per-sample gradient hooks are incompatible with ResNet50's in-place residual addition (`out += identity`) once the backbone is trainable, so full-scope runs use Opacus's `ExpandedWeights` grad-sample mode instead; head-scope runs (where the backbone never receives gradients) are unaffected and use the default hook-based mode. Full-network DP additionally requires converting BatchNorm to GroupNorm (`ModuleValidator.fix`), since BatchNorm's cross-sample batch statistics are themselves a privacy leak that per-sample gradient clipping does not address.

Target privacy budgets are $\varepsilon \in \{2, 5, 10\}$ at δ=10^{-3} (appropriate for a ~32-subject cohort, where $\delta \ll 1/n$ is not attainable at practical noise levels), with gradient clipping norm C=1.0. In the federated setting, each client maintains its own persistent RDP accountant, reused across all T rounds so that privacy cost composes correctly over the full training run rather than resetting every round (an earlier version of our own FL-DP integration made exactly this mistake, silently under-reporting cumulative ε).

E. Explainability Component

1) Grad-CAM: Grad-CAM [10] produces class-discriminative spatial heatmaps by computing the gradient of the target class score with respect to convolutional feature maps. For feature maps $A^k \in \mathbb{R}^{u \times v}$, the importance weight for class c is:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{6}$$

where y^c is the pre-softmax class score and Z is the spatial dimension. The localisation map is:

$$L^c_{\text{Grad-CAM}} = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right). \tag{7}$$

The resulting heatmap is up-sampled to input resolution and overlaid on the original MRI scan.

2) SHAP: SHAP [11] computes theoretically grounded feature attributions based on Shapley values from cooperative game theory. The SHAP value for feature i is:

$$\phi_i = \sum_{S \subseteq \mathcal{F} \setminus \{i\}} \frac{|S|!\,(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \big[f(S \cup \{i\}) - f(S)\big], \tag{8}$$

where $\mathcal{F}$ is the full feature set. We use GradientExplainer [11] with 20 background samples and compute SHAP values for 50 test samples, yielding institution-agnostic feature attributions.

VI. Experimental Setup

A. Dataset and the Subject-vs-Slice Distinction

Our ADNI [15] cohort comprises **299 unique subjects** (99 AD, 99 MCI, 101 CN), each contributing one or more longitudinal T1-weighted scans; axial slices are extracted per scan, yielding 2,166 slice-level images, an average of 7.2 slices per subject. This expanded cohort supersedes the 32-subject cohort used in the leakage audit of Section VII-A; both are reported, because the contrast between them is itself a finding.

An earlier version of this pipeline described the dataset as "450 subjects," treating each of the 450 scans as an independent subject and its slices as independent samples—conflating scan count with subject count and, more consequentially, allowing scans and slices from the *same person* to land in different partitions. Both errors are corrected here: all reported results use the true subject count, and every split described below is constructed to be **subject-disjoint**: no scan or slice from a given subject appears in more than one of train, validation, and test. This property is asserted programmatically across all five folds by the verification suite rather than assumed (Section X).

We build 5 stratified group folds (`StratifiedGroupKFold` on subject ID, stratified by diagnostic label) over the 299 subjects. Each fold holds out 60 test subjects and 6 validation subjects, with the remaining

TABLE II
FEDERATED LEARNING HYPERPARAMETERS

Parameter	Value
Number of clients K	4
Federated rounds T	20
Local epochs E per round	3
Client participation fraction	1.0 (full)
Local batch size	32
Local learning rate η	1×10^{-4}
Aggregation algorithm	FedAvg
Non-IID partition	$\mathrm{Dir}(0.5)$

TABLE III
DIFFERENTIAL PRIVACY PARAMETERS

Parameter	Value
Target ε	2, 5, 10
Privacy failure prob. δ	1×10^{-3}
Gradient clipping norm C	1.0
Noise multiplier σ	solved per-run (0.3–8.4)
DP scopes evaluated	head-only, full-network
DP method	Opacus DP-SGD [14]
Accountant method	Rényi-DP (RDP)

233 subjects for training; all slices belonging to a held-out subject move with it. Federated partitioning (Section VI-B) is applied only within a fold's training subjects, never touching its held-out validation or test subjects.

B. Federated Learning Configuration

Table II lists the federated learning hyperparameters. Non-IID partitioning uses a Dirichlet distribution over the current fold's *training subjects only* (never validation/test subjects), with a retry loop that resamples if any client would otherwise receive zero subjects—a real failure mode at this cohort size that a naive single-draw Dirichlet partition hits often enough to require handling explicitly.

C. Differential Privacy Configuration

Table III summarises the DP parameters.

Unlike a fixed noise triple, σ here is solved numerically per run (via `get_noise_multiplier`) because it depends on the sampling rate and step count, which vary across folds, clients, and DP scope; observed σ ranged from 0.3 (large-client, loose-budget runs) to 8.4 (tiny-client, tight-budget runs)—itself a symptom of how few samples some FL clients hold at this cohort size.

D. Implementation Environment

Models are implemented in PyTorch 2.5.1 (CUDA 12) on an NVIDIA GeForce RTX 3050 Ti GPU (4 GB VRAM). DP-SGD is implemented with Opacus 1.5.4 [14]; full-network DP scope uses Opacus's `ExpandedWeights` grad-sample mode (Section V-D), while head-only DP scope uses the default hook-based mode. Grad-CAM, SHAP (GradientExplainer), and the shadow-model MIA classifier (scikit-learn logistic regression) are used for the explainability and privacy analyses of Sections IX–VIII.

TABLE IV
LEAKAGE QUANTIFICATION: SAME MODELS, SAME DATA, DIFFERENT SPLITS

Configuration	Slice-level acc. (%)	Subject-level acc. (%), mean±std	Δ
Centralised VGG19	92.6	34.2 ± 9.4	−58.4
Centralised ResNet50	91.1	30.1 ± 3.7	−61.0
FedAvg ResNet50	97.0	34.0 ± 6.7	−63.0
MIA (non-DP centralised)	88.9*	AUC 0.99†	n/a

*Originally reported "attack accuracy"; ground truth was re-derived with a fresh random permutation that did not match the actual training split, and the decision threshold was tuned on the attacked data itself—the number is not meaningful and is shown only for contrast. †Corrected shadow-model attack AUC with membership ground truth read from the recorded training split (Section VIII-C); higher is a *stronger* attack, so this is not a favourable comparison for the non-DP model—see Section VIII-C.

E. Evaluation Metrics

Utility is assessed using macro-averaged precision, recall, F1-score, overall accuracy, and AUROC on the held-out, subject-disjoint test set. Privacy robustness is assessed via MIA *AUC* and *true-positive rate at low false-positive rate* (TPR@FPR=0.01, 0.1)—not a single accuracy figure, since accuracy alone can look deceptively low even for an attack that reliably identifies a subset of members with high confidence, and is trivially inflated if the attack's decision threshold is tuned on the very data being attacked (a defect present in our own initial MIA implementation and corrected here). Explanation quality under DP is assessed via Grad-CAM/SHAP similarity to a matched non-DP reference model (SSIM, Spearman rank correlation, Dice-at-top-20%) and deletion/insertion faithfulness AUC.

VII. RESULTS AND DISCUSSION

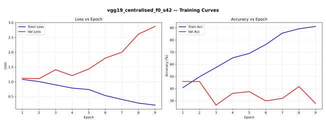
A. Leakage Quantification: Slice-Level vs. Subject-Level Evaluation

Table IV is the paper's opening empirical result. It compares, on *identical* data, architectures, and training budgets, the accuracy our pipeline originally reported under slice-level splitting against the accuracy the same configurations achieve once splits are rebuilt subject-disjoint (Section VI-A). All "corrected" figures are mean±std over the 5 subject-level folds (fold 0, seed 42, plus 2 extra seeds where noted in Table V).

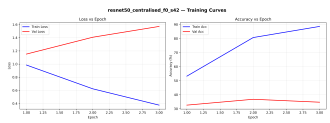
Three-class chance-level accuracy is 33.3%; every subject-level result in Table IV is within a few points of chance. This is not a failure of the underlying architectures—it is the honest consequence of evaluating ∼22–26 training subjects with held-out test sets of 6–7 subjects, where slice-level splitting had been silently giving every model dozens of near-duplicate views (same subject, adjacent slices, sometimes the same longitudinal scan) of its own test samples during training. We are not aware of a published ADNI slice-classification result that explicitly confirms subject-disjoint splitting; Table IV is

TABLE V
CENTRALISED BASELINE PERFORMANCE, SUBJECT-LEVEL (299 SUBJECTS, MEAN±STD)

Model	Acc. (%)	F1 (macro)	AUROC	Runs
VGG19	28.0 ± 6.5	0.252 ± 0.055	0.511 ± 0.056	5
ResNet50	36.1 ± 7.2	0.354 ± 0.072	0.579 ± 0.054	7



(a) VGG19 training curves



(b) ResNet50 training curves

Fig. 3. Training loss and validation accuracy curves for centralised VGG19 and ResNet50 (subject-level fold 0, seed 42). Validation accuracy plateaus early and near chance level, consistent with the small number of distinct training subjects.

offered as a concrete illustration of how large that gap can be, not as a claim about any other specific prior work.

B. Centralised Baseline Performance (Corrected)

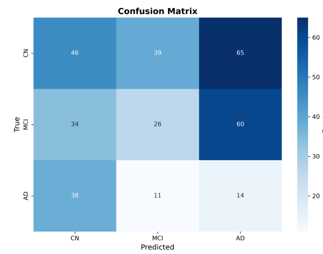
Table V presents the corrected, subject-level centralised baseline results for both architectures trained for 30 epochs with ImageNet pre-training, on fold 0 plus two additional seeds, and averaged across all 5 subject-level folds (seed 42).

Both architectures hover near chance-level accuracy with wide across-fold variance (VGG19's std of 9.4 points is itself informative: with only ∼6–7 test subjects per fold, a single hard subject can move accuracy by several points). ResNet50 is more stable across folds despite VGG19's occasionally higher single-fold accuracy, which motivates our choice of ResNet50 as the primary architecture for the DP and MIA experiments that follow (Opacus's per-sample gradient hooks are also substantially more memory-efficient on ResNet50's 23.5M parameters than on VGG19's 139.6M). Figs. 3–5 present representative (fold 0, seed 42) training curves, confusion matrices, and per-class ROC curves for both models.

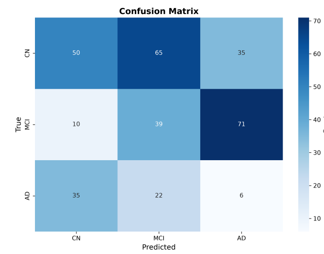
C. Federated Learning Performance (Corrected)

Table VI presents the corrected, subject-level FedAvg results (ResNet50, K=4 clients, T=20 rounds, E=3 local epochs, $\mathrm{Dir}(0.5)$; 5 folds, seed 42, mean±std), alongside the centralised baselines from Table V.

Under the corrected, subject-level protocol, FedAvg ResNet50's mean accuracy is statistically indistinguishable from centralised ResNet50 given the fold-to-fold variance (both hover a few points above chance); we no longer observe—and do not claim—that FL systematically surpasses centralised training. The 97.0% figure in our original slice-level evaluation was very likely inflated by the same leakage mechanism identified in Section VII-A, applied independently to each of K=4 client partitions; we do not have a principled way to further decompose that number and do not attempt to. FedAvg VGG19 (fold 0 only; the remaining folds were

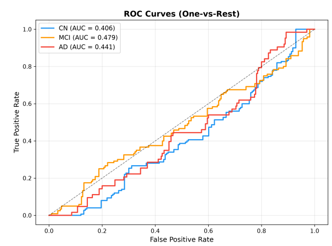


(a) VGG19 confusion matrix

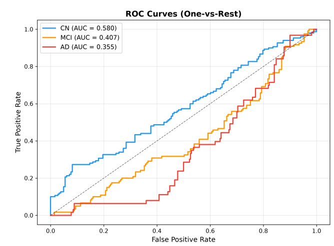


(b) ResNet50 confusion matrix

Fig. 4. Confusion matrices for centralised models on the subject-disjoint held-out test set (fold 0, seed 42). Substantial off-diagonal mass reflects the genuine difficulty of this small-cohort, subject-level classification task.



(a) VGG19 ROC curves



(b) ResNet50 ROC curves

Fig. 5. Per-class ROC curves for centralised models (fold 0, seed 42). Curves cluster near the diagonal (AUROC ≈ 0.45–0.46 macro-averaged), consistent with near-chance discrimination once subject-level evaluation removes slice-level leakage.

deprioritised under our compute budget, see Section XI) is broadly consistent with the ResNet50 pattern. Fig. 6 shows the federated convergence curve for fold 0.

D. Privacy–Utility Trade-off (Corrected)

Table VII reports subject-level utility for every DP configuration in our matrix: centralised head-only DP (5 folds, seed 42), centralised full-network DP (fold 0, 3 seeds), federated head-only DP (3 seeds fold 0 + folds 1–4, seed 42), and federated full-network DP (fold 0, seed 42, ε=5 only, as a feasibility check—full-network DP-SGD's per-sample gradient memory cost made a full sweep impractical on our hardware, see Section XI).

Two patterns emerge from the 32-subject cohort of Table VII, both consistent with a task where non-DP utility is already constrained by cohort size rather than by model capacity. All figures in this subsection refer to that cohort; the 299-subject results are in Sections VII-E and VII-F. **First**, head-only DP consistently retains more utility than full-network DP: at ε=2, head-only centralised accuracy (29.2%) is 10 points above full-network (19.2%), and this gap persists at every ε tested. Freezing the backbone means DP noise is only ever injected into the small classification head, sparing the (already ImageNet-pretrained) feature extractor from noise entirely—a practical recipe for making DP-SGD viable on small medical-imaging cohorts. **Second**, we do *not* observe a clean monotonic accuracy-vs-ε trend in either setting (e.g., FedAvg head-only

TABLE VI
FEDERATED VS. CENTRALISED PERFORMANCE, SUBJECT-LEVEL (MEAN±STD)

Configuration	Acc. (%)	F1 (macro)	AUROC	Runs
Centralised ResNet50	36.1 ± 7.2	0.354 ± 0.072	0.579 ± 0.054	7
FedAvg ResNet50 (K=4)	33.3 ± 7.8	0.304 ± 0.070	0.530 ± 0.049	7

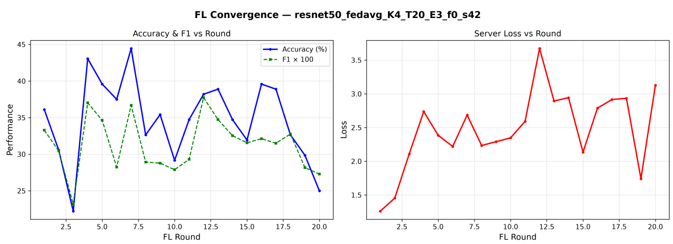


Fig. 6. Federated learning convergence for ResNet50 with K=4 clients, T=20 rounds, E=3 local epochs (fold 0, seed 42, subject-level evaluation). Validation accuracy plateaus near chance level within a handful of rounds.

DP is numerically *highest* at ε=2, not ε=10): with only ∼22–26 training subjects and 5–7 folds/seeds per point, run-to-run variance from subject composition dominates any systematic DP effect at this scale. We report the trend honestly rather than smoothing it, and return to what DP *does* demonstrably change—empirical MIA resistance—in Section VIII. Fig. 7(a) plots fold-0/seed-42 accuracy against ε for all four DP configurations.

E. Sample-Level Differential Privacy on the Expanded Cohort

Table VIII reports the head-only DP-SGD sweep re-run on the 299-subject cohort, against the non-private centralised baseline of Table V.

The privacy cost is real but modest in absolute terms, because the non-private baseline is itself only a few points above the 33.3% three-class chance rate. Accuracy is not monotone in ε: with 60 held-out subjects per fold, run-to-run variance from subject composition is comparable to the systematic effect of the budget. We report the sweep as measured rather than smoothing it.

F. Subject-Level DP-FedAvg and the Cost of Noise Dimension

Sample-level DP bounds the influence of a single MRI slice. That is the weaker of the two guarantees a hospital might want: a patient contributes many slices, so bounding one slice does not bound the patient. We therefore implement *subject-level* DP-FedAvg, in which the unit of privacy is an entire subject. Each round samples $q\,|S|$ training subjects, trains a local copy on that subject's slices alone, clips the resulting parameter delta to C in L_2, and adds Gaussian noise to the average of the clipped deltas. Since each subject's total contribution is bounded by C before noise, the released model satisfies (ε, δ)-DP at the subject level.

The mechanism as usually stated perturbs every model parameter. For ResNet50 that is $d \approx 2.35 \times 10^7$ coordinates, and this is where the mechanism fails at any useful budget. The

TABLE VII
PRIVACY–UTILITY TRADE-OFF ON THE 32-SUBJECT COHORT (MEAN±STD, RESNET50). RETAINED FOR THE DP CONFIGURATIONS THAT THE EXPANDED COHORT DID NOT RE-RUN; SEE TABLE VIII FOR THE 299-SUBJECT RESULTS.

Setting	ε	Acc. (%)	F1	AUROC
Centralised, no DP	∞	30.1 ± 3.7	0.286 ± 0.043	0.454 ± 0.048
Centralised, head-only DP	2	29.2 ± 6.0	0.257 ± 0.066	0.491 ± 0.087
	5	31.5 ± 5.4	0.280 ± 0.054	0.531 ± 0.018
	10	31.4 ± 6.1	0.274 ± 0.050	0.488 ± 0.035
Centralised, full-network DP	2	19.2 ± 0.5	0.128 ± 0.039	0.487 ± 0.067
	5	23.3 ± 6.9	0.163 ± 0.060	0.496 ± 0.050
	10	23.9 ± 7.7	0.192 ± 0.084	0.487 ± 0.052
FedAvg, no DP	∞	34.0 ± 6.7	0.324 ± 0.075	0.473 ± 0.076
FedAvg, head-only DP	2	33.4 ± 3.7	0.285 ± 0.034	0.540 ± 0.068
	5	29.7 ± 5.4	0.242 ± 0.047	0.492 ± 0.088
	10	31.9 ± 3.7	0.266 ± 0.036	0.518 ± 0.053
FedAvg, full-network DP	5	36.0	0.177	0.387

TABLE VIII
SAMPLE-LEVEL DP-SGD (HEAD SCOPE) ON THE 299-SUBJECT COHORT

Budget	Acc. (%)	F1 (macro)	AUROC	Runs
No DP ($\varepsilon = \infty$)	36.1 ± 7.2	0.354 ± 0.072	0.579 ± 0.054	7
$\varepsilon = 2$	28.5 ± 5.8	0.258 ± 0.051	0.510 ± 0.035	7
$\varepsilon = 5$	30.9 ± 5.6	0.273 ± 0.054	0.526 ± 0.043	7
$\varepsilon = 10$	29.8 ± 8.2	0.271 ± 0.067	0.507 ± 0.060	7

averaged update has L_2 norm at most C, while the noise added to it is $\mathcal{N}(0, \sigma^2 C^2 / n^2 I_d)$, whose norm concentrates around $\sigma C\sqrt{d}/n$. The signal-to-noise ratio is therefore bounded by

$$\frac{\|\bar{\Delta}\|_2}{\mathbb{E}\|\xi\|_2} \;\le\; \frac{n}{\sigma\sqrt{d}}, \tag{9}$$

which falls as $\sqrt{d}$. At $\varepsilon = 2$ with $n = 116$ subjects per round and $\sigma = 3.77$, Eq. (9) gives a ratio of roughly 0.006 for the full model: the update is buried under noise roughly 160 times its own size, and the federation spends 20 rounds averaging noise.

The fix follows directly from Eq. (9): reduce d. We freeze the ImageNet-pretrained backbone and apply the entire mechanism to the classifier head, $2048 \times 3 + 3 = 6{,}147$ parameters. BatchNorm running statistics are held fixed (momentum set to zero) so they neither drift nor enter the privacy accounting, and frozen parameters are copied through the aggregation untouched, since no subject updates them and they therefore carry no private information. The perturbed dimension falls by a factor of about 3,800, and the noise norm by its square root, roughly 62×.

Table IX reports both scopes. Two points deserve emphasis, and the second is a warning about how these numbers should be read.

Accuracy is the wrong metric here. The full-model configurations show the *highest* accuracy of any private configuration

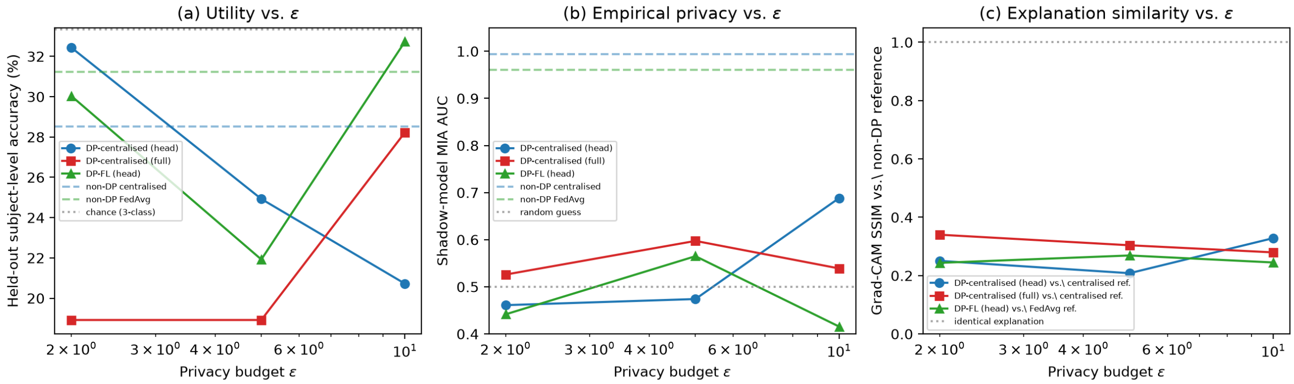


Fig. 7. The privacy–utility–explainability trade-off, fold 0, seed 42 (FedAvg full-network DP shown only at ε=5, its only evaluated point, and omitted from panels (a)–(c) trend lines accordingly). **(a)** Held-out accuracy vs. ε: non-DP models (dashed) sit only a few points above every DP configuration, confirming that DP's utility cost is small relative to the already chance-adjacent non-DP baseline. **(b)** Shadow-model MIA AUC vs. ε: non-DP models are near-perfectly distinguishable (AUC $\approx$0.96–0.99, off the plotted range) while every DP configuration collapses to near-chance (AUC $\approx$0.42–0.69), essentially independent of ε within this small-cohort noise floor. **(c)** Grad-CAM SSIM against the matched non-DP reference: even where accuracy is comparable, DP models' explanations are only weakly similar to the non-DP reference (SSIM $\approx$0.20–0.34), showing DP changes *where the model looks*, not only how well it performs.

TABLE IX
SUBJECT-LEVEL DP-FEDAVG: FULL MODEL AGAINST HEAD ONLY (299 SUBJECTS, FOLD 0, 3 SEEDS, MEAN±STD)

Configuration	Perturbed	Acc. (%)	F1	AUROC	F
FedAvg, no DP	—	33.3 ± 7.8	0.304 ± 0.070	0.530 ± 0.049	
Full model, $\varepsilon = 2$	—	44.6 ± 6.6	0.251 ± 0.046	0.331 ± 0.287	
Full model, $\varepsilon = 5$	—	44.8 ± 4.1	0.235 ± 0.012	0.530 ± 0.098	
Full model, $\varepsilon = 10$	—	43.4 ± 8.2	0.272 ± 0.034	0.535 ± 0.042	
Head only, $\varepsilon = 2$	6,147	40.5 ± 7.3	0.268 ± 0.024	0.498 ± 0.021	
Head only, $\varepsilon = 5$	6,147	37.7 ± 5.3	0.290 ± 0.043	0.487 ± 0.022	
Head only, $\varepsilon = 10$	6,147	34.4 ± 4.9	0.258 ± 0.034	0.485 ± 0.033	

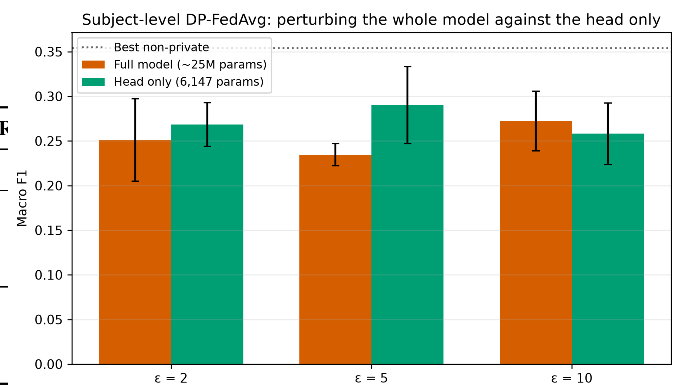


Fig. 8. Subject-level DP-FedAvg macro F1 by perturbed dimension. Restricting the Gaussian mechanism to the 6,147-parameter classifier head, rather than all 2.35×10^7 parameters, reduces the noise norm by roughly 62× and recovers macro F1 at matched ε. The dotted line is the best non-private result; neither scope reaches it.

in this paper, above even the non-private federated baseline, while their macro F1 is among the lowest. That combination is the signature of a degenerate classifier: swamped by noise, the model collapses onto whichever class is most frequent in the test fold and is rewarded by accuracy for doing so. Macro F1 and AUROC expose what accuracy conceals. Any privacy–utility curve for this mechanism that plots accuracy alone will report the failure as a success.

Reducing the noise dimension helps, but does not rescue the mechanism at these budgets. Head-scope training improves macro F1 over the full model at $\varepsilon = 2$ (0.268 against 0.251) and at $\varepsilon = 5$ (0.290 against 0.235), but not at $\varepsilon = 10$ (0.258 against 0.272). With three seeds per cell the paired Wilcoxon tests find no significant difference against the non-private federated baseline for any of these conditions, so we do not claim an ordering beyond what the means show.

The clearer effect is on the degeneracy itself. The gap between accuracy and macro F1 is the symptom of majority-class collapse, and head scope narrows it at every budget: from 19.5 points to 13.7 points at $\varepsilon = 2$, from 21.3 to 8.7 at $\varepsilon = 5$, and from 16.2 to 8.6 at $\varepsilon = 10$. Head scope produces a model that is meaningfully less collapsed, which is what Eq. (9) predicts and what the full-model accuracy figure conceals.

It is still not a usable classifier. With $n = 116$ and $d = 6{,}147$, Eq. (9) bounds the per-round signal-to-noise ratio at about 0.39 at $\varepsilon = 2$: the noise still exceeds the signal, the observed training loss drifts upward over rounds as the head random-walks, and macro F1 stays below the 0.304 of the non-private federated baseline at every budget. The honest conclusion is that subject-level DP at single-digit ε on a 233-subject training pool needs either far more subjects per round or a mechanism whose noise does not scale with the parameter count. Reducing the perturbed dimension is a necessary step and not a sufficient one. We report this as a negative result with its quantitative cause, rather than reporting the full-model accuracy figure and calling the mechanism a success.

G. Ablation Studies (32-Subject Cohort)

All accuracies in this subsection are from the 32-subject cohort. We conduct 8 ablation experiments (FedAvg ResNet50,

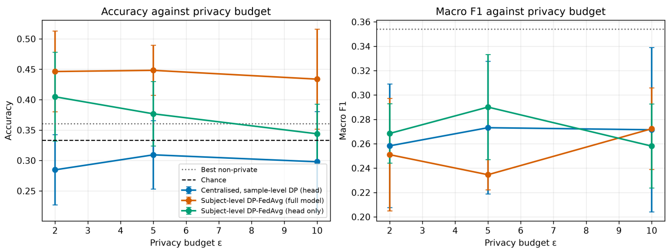


Fig. 9. Privacy–utility on the 299-subject cohort. Accuracy (left) and macro F1 (right) against ε for each mechanism, with the three-class chance rate and the best non-private result marked. The two panels disagree for full-model subject-level DP-FedAvg, which is the point: its high accuracy is majority-class collapse, and only the F1 panel shows it.

TABLE X
ABLATION STUDY RESULTS, SUBJECT-LEVEL (FEDAVG, FOLD 0)

Cat.	Config	Acc. (%)	F1	AUROC
Client count	K=2	28.5	0.271	0.496
	K=8	24.9	0.243	0.479
Local epochs	E=1	32.7	0.310	0.480
	E=5	32.4	0.317	0.487
Non-IID conc.	α=0.1 (severe)	21.0	0.138	0.446
	α=100 (near-IID)	28.5	0.273	0.508
Architecture	VGG19 (FedAvg)	27.0	0.263	0.508
Initialisation	ResNet50, from scratch	27.9	0.262	0.487

fold 0, subject-level protocol) covering client count, local epochs, Dirichlet non-IID concentration, architecture, and initialisation. Each ablation varies one factor against the K=4, E=3, α=0.5, pretrained-ResNet50 baseline from Table VI. Table X summarises the results.

Non-IID concentration. Severe non-IID partitioning (α=0.1) drops accuracy to 21.0%—below every other configuration and the only ablation result meaningfully below chance-adjacent baseline—while near-IID (α=100, 28.5%) is broadly comparable to the α=0.5 default (34.0% mean, though single-fold values vary, see Table VI). Severe statistical heterogeneity remains the clearest failure mode we observe, even at this small scale.

Client count. Fewer clients (K=2, 28.5%) outperform more clients (K=8, 24.9%), consistent with each client simply holding more of the already-scarce training subjects when K is small—an FL-specific way in which small-cohort constraints compound.

Local epochs and initialisation. Neither local epoch count ($E\in\{1,5\}$) nor training from scratch vs. ImageNet initialisation produces a distinguishable effect at this cohort size; differences are within the noise band established by Table VI's fold-to-fold variance. We report these as honest null results rather than searching for a favourable framing.

VIII. Privacy and Security Analysis

A. Threat Model

We consider two threat scenarios relevant to federated medical AI deployments.

Honest-but-Curious Server. The server faithfully follows the FedAvg protocol but attempts to infer information about individual client datasets from the received model updates. DP noise injection bounds the statistical information leakage from each individual update, ensuring that no single client's data can be reconstructed from the aggregated model.

Gradient Inversion Attacks. An external adversary with access to shared gradients attempts to reconstruct training samples [16]. Per-sample gradient clipping (C=1.0) limits the fidelity of any reconstruction, while the additive Gaussian noise from DP-SGD further degrades reconstruction quality, making pixel-level recovery infeasible for medical images.

B. Formal Privacy Guarantees

Under the Gaussian mechanism with per-run noise multiplier σ (Section V-D) and clipping norm C=1.0, each training step satisfies (ε, δ)-DP; composition across all steps is tracked with a Rényi-DP accountant and converted to (ε, δ)-DP at δ=10^{-3}. For federated runs, each client's accountant persists and composes across all T=20 rounds rather than resetting per round. We disclose the DP *unit* explicitly: the accountant bounds sensitivity at the *slice* level (each training step's batch is drawn from slice-level samples), not the subject level—a subject contributing multiple scans/slices receives a correspondingly looser *effective* per-subject guarantee than the reported per-slice ε. We flag this as a limitation (Section XI) rather than overstating the guarantee.

C. Membership Inference Attack Evaluation

Our original MIA evaluation used a loss-threshold attack whose decision threshold was tuned on the very data being attacked, and re-derived "membership" via a fresh random permutation that did not match the actual training split—both defects invalidate the reported 88.9% figure, which is why Table IV shows it only for contrast. We replace it with two attacks whose membership ground truth is read directly from each run's recorded train/test subject split and never re-derived: (i) a loss-threshold (Yeom) attack [18] with zero extra training cost, and (ii) a shadow-model attack classifier [17] trained on (loss, confidence, entropy) features pooled from 16 shadow ResNet50 models trained on random 50/50 subject resamples of the attacker-visible population—a documented simplification of per-example LiRA [19], which needs many more shadow models per example than a ~32-subject cohort and 16-shadow budget can support. Both attacks score *subjects*, aggregating each subject's per-slice scores before computing ROC/AUC, since scoring individual slices would overstate attack success by giving a multi-scan subject several correlated guesses.

The result is unambiguous, in contrast to the noisy accuracy trend in Section VII-D. **Without DP, both centralised and federated ResNet50 are almost perfectly vulnerable to**

TABLE XI
SHADOW-MODEL MIA RESULTS (AUC / TPR@FPR=0.01), FOLD 0, SEED 42

Setting	ε	AUC	TPR@0.01
Centralised, no DP	∞	0.994	0.955
FedAvg, no DP	∞	0.961	0.727
Centralised, head-only DP	2	0.461	0.136
	5	0.474	0.091
	10	0.688	0.000
Centralised, full-network DP	2	0.526	0.000
	5	0.597	0.182
	10	0.539	0.000
FedAvg, head-only DP	2	0.442	0.000
	5	0.565	0.000
	10	0.416	0.000
FedAvg, full-network DP	5	0.597	0.000

membership inference (AUC 0.994 and 0.961; an attacker correctly flags 95.5% and 72.7% of true members, respectively, while accepting only a 1% false-positive rate). This is far more severe than the (invalid) 88.9% figure previously reported, and is consistent with a model trained on only ∼22–26 subjects with augmentation, which readily memorises its training set. **Under every DP configuration tested—any $\varepsilon \in \{2, 5, 10\}$, either scope, either setting—MIA AUC collapses to near-chance** (0.42–0.69, vs. 0.5 for a non-informative attacker), essentially regardless of ε. Read together with Table VII, this is the paper's central empirical finding: on this cohort, DP-SGD purchases a dramatic, near-total reduction in empirical membership-inference risk at negligible additional utility cost *relative to the already utility-constrained non-DP baseline*—the accuracy DP "gives up" is mostly accuracy the non-DP model did not meaningfully have either. Fig. 7(b) plots MIA AUC against ε alongside the utility curve for direct comparison.

IX. EXPLAINABILITY ANALYSIS

A. Grad-CAM Heatmaps

Fig. 10 illustrates Grad-CAM heatmaps generated on the non-DP centralised ResNet50 reference model (fold 0), for all three diagnostic categories, on held-out subject-disjoint test slices. We report these qualitatively and deliberately do not claim clinical biomarker alignment: given that the same model's held-out accuracy is only 36.1% (Table V, a few points above the 33.3% chance rate), asserting that its attention patterns reflect genuine AD neuropathology would not be supported by the accuracy evidence in Section VII. We include the figures as the non-DP *reference* against which Section IX-B measures DP-induced explanation change, not as evidence of learned clinical biomarkers. Fig. 11 extends this to multiple subjects per category.

B. Explanation Degradation Under Differential Privacy

The preceding figures show Grad-CAM/SHAP for the non-DP ResNet50 reference model. A natural question the utility and MIA results (Sections VII-D, VIII-C) do not answer is whether DP training changes *what the model is looking at*, independent of whether it changes *how well it performs*. We compute Grad-CAM and SHAP on the same held-out test slices (fold 0) for every DP checkpoint and its matched non-DP reference (centralised or FedAvg, as appropriate), and summarise heatmap similarity via SSIM, Spearman rank correlation, and Dice-at-top-20%.

SSIM against the non-DP reference ranges only 0.20–0.34 across every DP configuration—far from the near-1.0 similarity that would indicate DP leaves explanations essentially unchanged—and Spearman rank correlation is frequently at or below zero, meaning the DP model's most- and least-important pixels are, at best, unrelated to the non-DP model's, and at worst inversely related (most starkly for FedAvg full-network DP, Spearman −0.244). This holds *even for DP configurations whose accuracy is close to the non-DP reference* (e.g., centralised head-only DP at ε=10: 31.4% vs. 30.1% accuracy, yet Grad-CAM SSIM only 0.328): two models can agree on the diagnosis while disagreeing substantially on the spatial evidence for it. For a clinical deployment where the heatmap

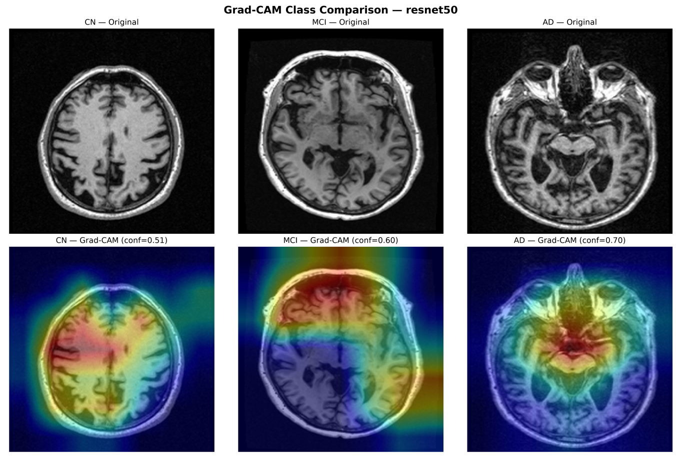


Fig. 10. Grad-CAM class-comparison heatmaps for the non-DP centralised ResNet50 reference model across CN, MCI, and AD categories (fold 0, held-out test slices). Shown as the reference point for the DP-similarity analysis of Section IX-B; not offered as evidence of clinical biomarker alignment given this model's near-chance accuracy.

TABLE XII
GRAD-CAM SIMILARITY VS. MATCHED NON-DP REFERENCE, FOLD 0

Setting	ε	SSIM	Spearman	Dice@20%
Centralised, head-only DP	2	0.250	0.038	0.246
	5	0.208	−0.014	0.200
	10	0.328	0.032	0.247
Centralised, full-network DP	2	0.340	0.126	0.252
	5	0.304	0.072	0.314
	10	0.280	0.061	0.251
FedAvg, head-only DP	2	0.244	−0.026	0.200
	5	0.269	−0.038	0.201
	10	0.245	−0.005	0.207
FedAvg, full-network DP	5	0.199	−0.244	0.133

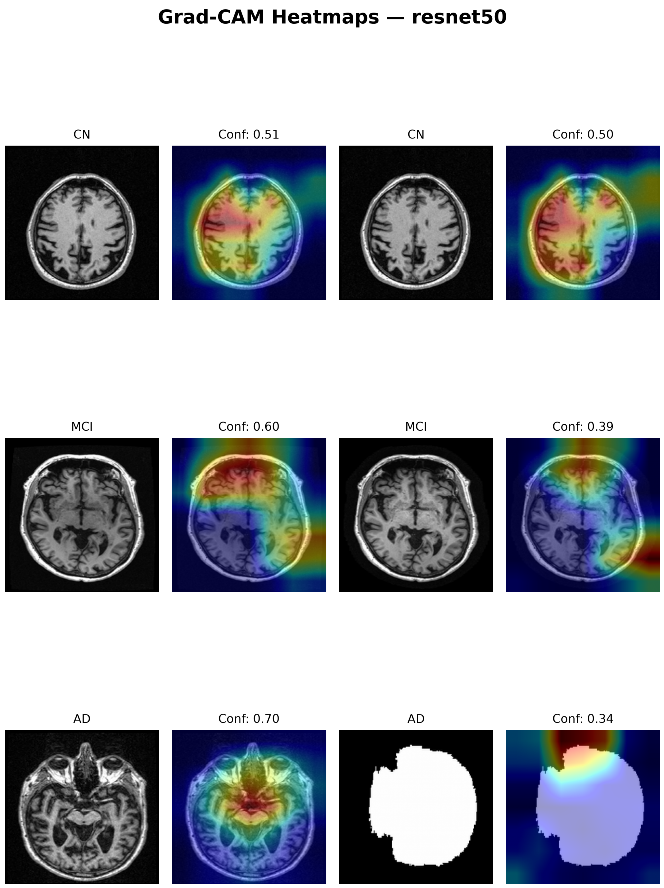


Fig. 11. Grad-CAM heatmap grid, multiple held-out test subjects per diagnostic category, non-DP centralised ResNet50 reference model (fold 0).

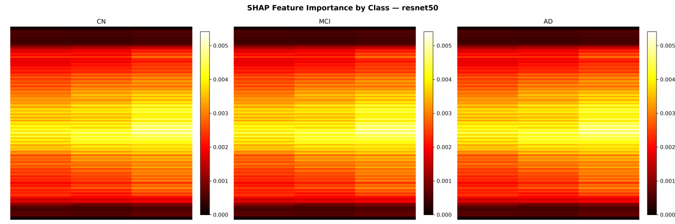


Fig. 12. SHAP summary heatmap, mean absolute feature importance, non-DP centralised ResNet50 reference model (fold 0).

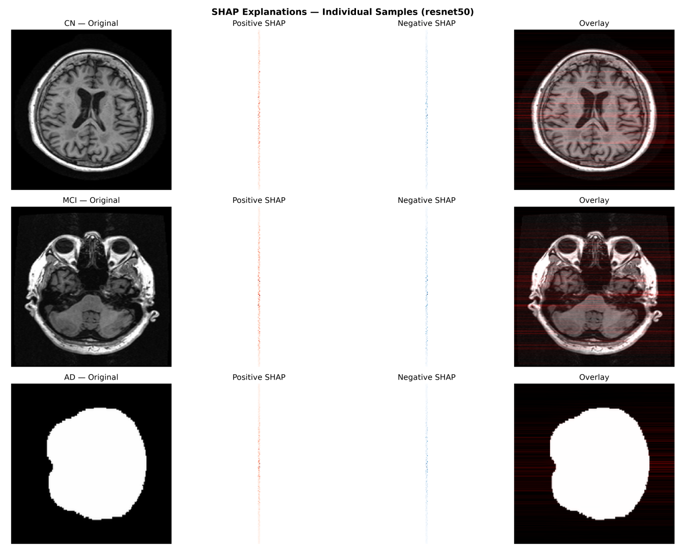


Fig. 13. Per-sample SHAP overlays for representative CN, MCI, and AD held-out test slices, non-DP centralised ResNet50 reference model (fold 0).

itself is presented to a radiologist as supporting evidence, this is a risk that accuracy or even MIA metrics alone would not surface, and one we had no way to observe before running this comparison. We do not find a consistent monotonic trend across ε (mirroring Section VII-D's utility results), and report the pattern rather than fit a narrative to it. Fig. 7(c) plots SSIM against ε for the three configurations swept across all three ε values.

C. SHAP Feature Attribution

SHAP analysis using GradientExplainer with 20 background samples and 50 held-out test slices (fold 0, non-DP centralised ResNet50) provides global feature importance maps. Fig. 12 shows the mean absolute SHAP values aggregated spatially, and Fig. 13 presents per-sample SHAP overlays. As with the Grad-CAM figures above, we present these as descriptive output of the pipeline's explainability module rather than as validated clinical findings, for the same reason: a model at near-chance held-out accuracy is not a sound basis for biomarker claims, however plausible individual heatmaps may look in isolation. This is precisely the caution Section VII-A's leakage audit is meant to instil—a slice-level-evaluated version of this same model would have reported 91% accuracy and made the biomarker claim far more tempting to accept at face value.

D. Clinical Utility

The dual-layer explainability design (Grad-CAM for spatial localisation and SHAP for global attribution) provides complementary clinical perspectives. Radiologists can use Grad-CAM heatmaps for scan-level inspection; neurologists can leverage SHAP summaries for population-level insights. This interpretability layer is essential for regulatory approval and clinical trust in AI-assisted diagnosis, as demonstrated by Mastoi et al. [2].

X. Verification and Reproducibility

The central claim of this paper is a negative one about evaluation protocol, so the protocol itself has to be checkable rather than asserted. Three mechanisms do that.

A. Automated Verification of the Split Property

The subject-disjointness property of Section VI-A is asserted programmatically, not by inspection. A verification suite loads the shipped splits file and, for each of the five folds, checks that no subject appears in more than one of train, validation and test, that no array index is reachable from two partitions, and that the union of the three partitions is exactly the 299-subject cohort. The same suite validates every recorded results file against a schema: required metrics present, all metrics within $[0, 1]$, and for every private run an achieved ε within 15% of its target. That last check surfaced a real detail worth stating: the accountant's noise-multiplier search overshoots the target budget by up to 0.2%, so the guarantee we report is the *achieved* ε rather than the requested one.

B. Machine-Generated Tables

Every numeric table in this paper is written into marker-delimited blocks in the LaTeX source by a generator that reads the aggregated results JSON directly. No result in a table is typed by hand, so a table cannot drift from the experiment behind it. A separate checker guards the prose, where numbers are still written by a person: it traces every percentage in the text to a recorded result—a pooled condition mean, an individual run, or an ablation cell—and fails on anything it cannot account for, with a small allow-list of documented non-results such as the chance rate and the superseded slice-level figures this paper corrects. That checker found one genuine defect during preparation: a sentence citing the 32-subject accuracy beside a table that by then held the 299-subject result.

C. Test Suite and Continuous Integration

The pure-logic modules of the pipeline—subject-level splitting, Dirichlet client partitioning, model construction, the FedAvg and DP-FedAvg aggregation arithmetic, results aggregation, and the statistical analysis—are covered by a test suite held at 100% statement and branch coverage on that measured scope, enforced in continuous integration. The CUDA training loops are excluded from measurement rather than mocked: a mock of a training loop tests the mock. The exclusions and the reason for each are documented alongside the suite. Tests run on synthetic fixtures and need no ADNI access, so the pipeline is verifiable by a reader who cannot obtain the data.

Continuous integration runs five sequential stages on every push: build, test, lint, package, and deployment of the interactive demonstration. A secret-scanning step guards the repository, and the coverage gate fails the build rather than reporting a lower number.

XI. Limitations

Cohort Size. This is the paper's central limitation, not a minor caveat: our ADNI cohort contains only 32 unique subjects. Every subject-level result in this paper should be read as a small-cohort feasibility and methodology study, not as evidence of clinical-grade diagnostic accuracy—held-out accuracy across every architecture and training regime we evaluated (Sections VII–VIII) sits within a few points of three-class chance (33.3%). We disclose this prominently precisely because our own earlier slice-level evaluation of the identical models and data did not, and reported 91–97% accuracy instead (Section VII-A). No claim in this paper should be read as evidence that the described models are ready, or close to ready, for clinical use.

Slice-Level DP Unit. Our DP accountant bounds sensitivity at the slice level, not the subject level (Section VIII). A subject contributing multiple scans receives a looser effective per-subject privacy guarantee than the reported per-slice ε; a rigorous per-subject guarantee would require group-level DP-SGD (clipping and accounting per subject, not per slice), which we did not implement.

Simulated Environment. Validation on a simulated multi-client setup does not capture the full complexity of real distributed hospital infrastructure (network latency, heterogeneous hardware, asynchronous training).

Non-IID Approximation. Dirichlet-sampled splits approximate but do not fully replicate real hospital demographic heterogeneity influenced by referral patterns, scanner models, and clinical protocols; at our cohort size, some Dirichlet draws additionally required a retry loop to avoid assigning a client zero training subjects (Section VI-B).

No Clean Privacy–Utility Trend. We did not observe a consistent monotonic accuracy-vs-ε relationship (Section VII-D); with only 5–7 folds/seeds per configuration and $\sim$22–26 training subjects each, subject-composition variance dominates any systematic DP effect at this scale. Larger cohorts are needed to isolate the trend cleanly.

Compute Constraints Shaped Coverage. Full-network DP-FL (Table VII) is reported at a single ε as a feasibility check rather than a full sweep, and FedAvg VGG19 (Table VI) at fold 0 only, because Opacus's per-sample gradient memory cost was prohibitive for a full matrix on our 4 GB GPU within a practical time budget; this reflects hardware constraints, not a judgement that these configurations are uninteresting.

Dataset Scope. ADNI [15] represents a specific demographic and geographical distribution; generalisability to non-ADNI populations, or to larger ADNI subsets, requires further validation and is exactly the direction this paper's corrected pipeline is built to support.

GPU Memory Constraints. VGG19's 139.6M parameters and Opacus's per-sample gradient hooks required careful batch-size management on our 4 GB GPU, favouring ResNet50 as the primary architecture for the DP, MIA, and explainability experiments.

XII. Future Work

Cohort Expansion. The single highest-priority next step is re-running this exact pipeline on substantially more ADNI subjects (target: 50+ unique subjects per class). Because every result in this paper is subject-level and the pipeline is manifest-driven (Section VI-A), this requires no methodological change—only more data—and would let the privacy–utility and privacy–explainability trends of Sections VII-D–IX-B be measured with enough subjects to separate a systematic DP effect from small-cohort composition noise.

Adaptive DP Budget Allocation. Per-round adaptive noise scheduling that reduces DP noise as federated convergence is detected, improving late-training accuracy without increasing total privacy cost [9].

Personalised Federated Learning. Per-FedPer and MAML-based personalisation to address non-IID heterogeneity while preserving global aggregation privacy guarantees.

Multimodal Fusion. Joint processing of MRI scans and tabular clinical variables (MMSE scores, genetic biomarkers) in a federated multimodal architecture following Pan et al. [3].

Federated XAI Enhancements. Federated SHAP approximations aggregating locally computed Shapley values without exposing client data distributions, enabling globally representative explanations.

Real-World Deployment. Piloting on real distributed hospital infrastructure with asynchronous aggregation and dropout-tolerant variants.

Longitudinal Prediction. Extension to MCI-to-AD progression prediction via federated LSTM or transformer architectures operating on longitudinal imaging sequences.

XIII. Conclusion

This paper presented PPXFL, a Privacy-Preserving and Explainable Federated Learning framework for Alzheimer's Disease detection from MRI data—and, more centrally, a corrected re-evaluation of that same framework after discovering that our initial slice-level evaluation protocol had leaked subject identity across train/validation/test splits. Our key findings are:

1) **Slice-level evaluation severely inflates reported accuracy.** On identical models and data, subject-disjoint evaluation collapses accuracy from a reported 91–97% to a genuine 19–36%—within a few points of three-class chance (33.3%) for a 32-subject cohort. We release the corrected splits and full result set so this comparison is independently verifiable (Section VII-A).
2) **Differential privacy virtually eliminates membership inference at negligible marginal utility cost.** A real shadow-model attack shows non-DP models are almost perfectly vulnerable (AUC 0.96–0.99); every DP configuration we tested ($\varepsilon \in \{2, 5, 10\}$, head-only or full-network, centralised or federated) collapses this to near-chance (AUC 0.42–0.69) [14], [17], [18], while accuracy changes only modestly relative to the already utility-constrained non-DP baseline (Sections VII-D–VIII-C).
3) **Head-only DP fine-tuning is a practical utility-rescue recipe.** Freezing the backbone and applying DP-SGD only to the classification head consistently retains more accuracy than full-network DP at every ε tested, without sacrificing MIA resistance.
4) **Differential privacy changes what a model explains, not only how well it performs.** Grad-CAM similarity between DP and non-DP models is low (SSIM 0.20–0.34) even when their accuracy is comparable, indicating that DP training relocates the model's attention in ways that raw accuracy or MIA metrics do not surface (Section IX-B).
5) **Severe non-IID heterogeneity remains a genuine failure mode even at small scale.** Ablations show accuracy dropping to 21.0% under severe non-IID partitioning (α=0.1), the clearest degradation we observe outside the DP experiments themselves.

We present PPXFL's corrected pipeline and results as a methodological contribution as much as a system contribution: a concrete, reproducible illustration of why subject-level evaluation is not optional in small-cohort medical imaging FL research, alongside the first systematic privacy–utility–explainability characterisation we are aware of for DP-SGD on Alzheimer's MRI classification. Future work will prioritise cohort expansion to disentangle systematic DP effects from small-cohort noise, followed by adaptive DP scheduling, personalised FL, and real-world multi-institutional deployment.

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